

Aliah University

End-Semester Examination (Autumn Semester) - 2023

(For 2nd Year 3rd Semester B.Tech (ECE))

Paper Name: Data Structures and Algorithms Analysis
Paper Code: CSEUGOE01

Full Marks: 80
Time: 3 hrs

Group-A

(Answer the following question by choosing the correct answer from the given options)

10 × 1 = 10

1. A data structure that contains data and a pointer to the next object is a
(a) Template (b) class (c) pointer class (d) linked list
2. In a stack, the data item placed on the stack first is,
(a) The first data item to be removed (b) The last data item to be removed
(c) Given index as zero (d) Not given as index number
3. Link list are not suitable for
(a) Insertion Sort (b) Binary Search (c) Radix Sort (d) Polynomial function
4. Stack is useful for implementing
(a) Radix Sort (b) Breadth first Search (c) Recursion (d) None of these
5. We use malloc and calloc for
(a) Dynamic memory allocation (b) static memory allocation
(c) both dynamic and static memory allocation (d) None of these
6. The maximum number of nodes in a binary tree of height k is
(a) $2^{k+1} - 1$ (b) $2^k - 1$ (c) $2^k + 1$ (d) 2^{k+1}
7. Number of comparison in searching an element in a binary search tree is
(a) $O(n)$ (b) $O(\log n)$ (c) $O(n^3)$ (d) $O(n \log n)$
8. Bubble sort and quick sort belong to which of the following category of the algorithm
(a) $O(n \log n)$ sorts (b) Divide and conquer sorting
(c) Average case quadratic (d) Exchange sort
9. Which of the following probing methods suffer from the problem of secondary clustering?
(a) linear probing (b) Quadratic probing (c) double hashing (d) both (b) and (c)
10. If the array is sorted, then the time complexity will be
(a) $O(n)$ (b) $O(n \log_2 n)$ (c) $O(n^2)$ (d) $O(\log_2 n)$

Group-B

(Answer any six questions)

6 × 5 = 30

1. (A) What are the different types of data structures? (B) Write down the four applications of data structure? (C) Write down the algorithm for solving Towers of Hanoi problem in brief? (1+2+2)
2. (A) What is a Stack? (B) What are the two operations in Stack? (1+1+3)
(C) Consider the following stack of characters, where STACK is allocated N=10 memory cells. STACK: A, C, D, F, K, _, _, _
Describe the stack as the following operations take place:
i) POP (STACK, ITEM)
ii) PUSH (STACK, L)
iii) PUSH (STACK, P)
3. (A) Define a linked list. (B) Write and explain the algorithm of inserting an element at the end of the link list. (C) What are the operations in a singly linked list? (1+3+1)
4. (A) Simulate the Merge sort algorithm and show the step-by-step procedure with given data values: 23, 11, 37, 28, 15, 19, 55. (B) Write the worst case complexity of merge sort? (4+1)

5. Write an algorithm for postfix expression, evaluate it and show the contents of stack for the following postfix expression:
 $P: 312 \div - 241 / + * 153 +$
 $(2+3=5)$
 (Note: $\wedge$ stands for power and all operands are single digit). (3+2)
6. The inorder and preorder traversal of a tree are given below:
 Inorder: D B M I N E A F C J G K
 Preorder: A B D E I M N C F G J K
 A) Construct the corresponding Binary Tree
 B) Determine the postorder traversal of the tree (3+2)
7. (A) Define Graph as a data structure? How to represent a graph?
 (B) What are the applications of graphs? (3+2)

Group-C

(Answer any four questions)

$4 \times 10 = 40$

1. (a) What is a Binary Search Tree (BST)? Construct a BST for the following numbers: 47, 55, 23, 17, 39, 11, 50, 9, 19, 71, 33, and 28. Show all the steps.
 (b) Write its preorder traversal. (8+2)
2. A) Consider the following arithmetic infix expression Q:
 $Q: A = (B * C + (D * E \wedge F) / G) * H$
 Convert this expression into equivalent postfix expression P and show stack contents. (8+2)
 B) Write the benefits of link list over Array?
3. (A) Define Hash function? (2+3+5)
 (b) What are the methods of resolving collisions in the hashing?
 (c) Consider the following 4 digit employee numbers
 9614, 5882, 6713, 4409, 1825
 Find the 2 digit hash address of each of the number using the division method with $m=97$
4. A) Simulate the Binary Search algorithm and show the step-by-step procedure with given data values: 11, 22, 30, 33, 40, 44, 55, 60, 66, 77, 80, 88, and 99. (6+4)
 B) Explain Quick sort algorithm with an example?
4. (a) Define AVL Tree? (b) Write the data structure for the AVL Tree? (c) Show the sequence of modification in the AVL tree (initially empty) on inserting the sequence of keys 30, 12, 45, 18, 10, 15, 40, 20, 22. (d) write the number of different rotations required for the normalization? (1+2+5+2)
5. Write notes on any two (5+5)
 (a) Adjacency matrix
 (b) Breadth first search
 (c) Hashing

B.Tech. Examination-2023-24
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)

Course Title: Physics of Semiconductor Devices

Course Code: ECEUGPC03

Full Marks: 80

Time: 3.00 Hrs

		Marks
1. (a)	Discuss the relation between the Heisenberg's uncertainty principles and classical wave dynamics.	5
(b)	Starting from Schrödinger equation, derive an expression of Fermi-Dirac probability of occupancy of the n^{th} quantum state of an electron in an one-dimensional infinite quantum well of width ' a '.	11
2. (a)	What is relaxation time? What is mean free path? What is Matthiessen's rule?	3x2
(b)	Discuss the dependence of the bulk mobility of charged carriers in a semiconductor on temperature?	5
(c)	Let the mean time of flights of two separate scattering mechanisms are τ_1 and τ_2 . Let the corresponding mean free paths are l_1 and l_2 . Let the velocity of electron between two successive scattering events is v_{th} . Prove that the effective mean time of flight and the mean free path are given as respectively, $1/\tau = 1/\tau_1 + 1/\tau_2$, and $1/l = 1/l_1 + 1/l_2$.	5
3. (a)	Derive the expressions of the electric field and the electrostatic potential in the space charge region of a p-n junction. Assume all necessary parameters.	10
(b)	A silicon p-n junction is to be built in an n-type bulk region having $N_d = 10^{15} \text{ cm}^{-3}$, by <i>compensating</i> with a larger density of acceptor impurities (N_a). Optimize the value of N_a to achieve a junction capacitance per unit area of the junction of 3.66 nF/cm^2 at a reverse bias of 5 V. Assume the $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.	6
4.	A silicon p-i junction is created between a p-type region ($N_a = 10^{18} \text{ cm}^{-3}$) and an intrinsic region.	
(i)	Draw the equilibrium and forward bias ($V_F = 0.2 \text{ V}$) band diagrams of the junction. Mark the numerical values of all important parameters on the band diagrams.	6
(ii)	Draw the qualitative distributions of the excess/deficient carrier densities in both neutral regions under forward bias condition.	4
(iii)	Along with diagram of (ii), show all the significant current components of the ideal forward bias current of the p-i junction under forward bias.	3
(iv)	Calculate the width of the space charge region of the p-i junction at a reverse bias of $V_R = 2.0 \text{ V}$.	3
5.	Derive the expressions of the steady state minority carrier distributions in the neutral regions of an n-p-n Bipolar Junction Transistor biased in forward active operating mode. Define all necessary quantities.	16
6. (a)	Explain the optimization of the emitter injection efficiency of a Heterojunction Bipolar Transistor.	10
(b)	Design the base of an n-p-n BJT for the base transport factor of 0.9880. In the base region, the electron diffusion constant is $25 \text{ cm}^2 \cdot \text{s}^{-1}$ and the excess electron life time is $\tau_{n0} = 4 \times 10^{-7} \text{ s}$. Calculate the required base width.	6
7. (a)	Draw the band diagram of a NMOS transistor along source-channel-drain direction when it is under following conditions. Always, $V_{SB} = 0$. Assume threshold voltage is	8

quite small.

(i) $V_{GS} = 0, V_{DS} = 0$, (ii) $V_{GS} = 0, V_{DS} = V_{DD}$, (iii) $V_{GS} = V_{DD}, V_{DS} = 0$, (iv) $V_{GS} = V_{DD}, V_{DS} = V_{DD}$.

- (b) What is channel length modulation effect? What is punch-through condition? 4
- (c) Consider a bulk silicon NMOS fabrication process with $L_{min} = 0.18 \mu m$, $\mu_n = 650 \text{ cm}^2/\text{V.s}$, $V_{T0} = 0.50 \text{ V}$, $t_{ox} = 10 \text{ nm}$, $\epsilon_{SiO_2} = 3.9\epsilon_0$. Design an minimum area NMOS transistor to drive a saturation drain current of $I_{DSAT} = 2 \text{ mA}$, when $V_{GS} = 1.8 \text{ V}$. 4
8. (a) Derive an expression of the threshold voltage change due to shortening of the channel length of an NMOS transistor. 10
- (b) How are the MOSFET source/drain diffusion regions are optimized to mitigate hot-electron effect? What is LDD structure in MOSFET device architecture? 6

$$I_{DS} = \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{th} - V_{T0}) V_{DS} - \frac{V_{DS}^2}{2}$$

-End-

$$n_{p_0} = \frac{P_{p_0}}{kT}$$

$$n_{p_0} = N_{Cn} \left(\frac{-E_{Fn} - E_{Cn}}{kT} \right)$$

$$n_{n_0} = N_{Cn}$$

B.Tech. Examination-2023-24
Electronics and Communication Engineering Department
 (Odd Semester Regular and Supplementary)
 Course Title - Analog Electronics Course Code -ECEUGPC01

Full Marks : 80

Time-3 hrs

- (Answer any five, Symbols have their usual meaning)
1. (a) Show that saturation region drain current expression in n-channel MOSFET is $I_D = \frac{1}{2} \mu_n C_{OX} \frac{W}{L} (V_{GS} - V_{TH})^2$ 6
- (b) Determine the Q-point (V_{CEQ}, I_{CQ}, I_{BQ}) in emitter bias Si-BJT circuit having $V_{CC} = 10V$, $R_1 = 5.6k\Omega$, $R_2 = 1k\Omega$, $R_C = 4.7k\Omega$, $R_E = 1k\Omega$ & $\beta = 10$. Draw the load line and Q-point of the circuit. 7
- (c) Calculate the Early voltage of bipolar transistor having output resistance $r_o = 225k\Omega$ at $I_{CQ} = 0.8 mA$. 3
- OR
1. (a) Explain the voltage divider biasing scheme of BJT with proper circuit diagram and explain why this biasing arrangement is preferred? 6
- (b) Derive the expression of stability factor with respect to I_{CO} . 4
- (c) Given that $V_{CC} = 18V$, $V_{CEQ} = 10V$, $I_{CQ} = 2 mA$, $R_1 = 18k\Omega$, $R_E = 1.2k\Omega$ for a common emitter configuration-based voltage divider biasing network. Determine the values of R_2 and R_C . (Draw the appropriate circuit diagram) 6
2. (a) Show that small signal voltage gain of the circuit shown in fig. 1 is $A_v = -(g_m R_C) \left(\frac{r_\pi}{R_B + r_\pi} \right)$ 6
- (b) Establish the relation $f_T = \beta_o f_\beta$ (unity-gain bandwidth) of BJT after estimating the short-circuit current gain of hybrid- π equivalent model of BJT. 6
- (c) Determine the 3 dB cutoff frequency of the short-circuit current gain of a bipolar transistor with parameters $r_\pi = 2.6k\Omega$, $C_\pi = 0.5pF$ & $C_\mu = 0.025pF$. 4
- OR
2. (a) Draw the common emitter configuration based fixed bias network for the ac response with the r_e model inserted between the appropriate terminals by including r_o and derive the expression of Z_i , Z_o and A_v . 8
- (b) A fixed bias network based on CE configuration has the following specifications such as $V_{CC} = 10V$, $R_B = 470k\Omega$, $R_C = 3k\Omega$, $\beta = 100$ and $C_1 = C_2 = 10\mu F$. For this network determine the following parameters using the r_e model of transistor: (i) r_e (ii) Z_i (with $r_o = \infty$) (iii) Z_o (with $r_o = \infty$) (iv) A_v 8
3. (a) Draw the Wien Bridge oscillator circuit and show that the gain of the amplifier $A = 3$ and the frequency of oscillation $f_0 = \frac{1}{2\pi RC}$, in sustain oscillation condition. 2+4+4 = 6
- (b) Determine the frequency of oscillation of the circuit shown in Fig. 2. 1+5 = 6
4. (a) Identify the oscillator circuit and estimate the frequency of oscillation shown in Fig. 3. 3
- (b) What is the gain of the amplifier in sustain oscillation condition of the oscillator? 2+2 = 4
- (c) Derive the expression of series and parallel resonant frequency of quartz crystal. 3
- (d) Draw the parallel resonant mode of crystal oscillator circuit. 5
5. (a) Draw the circuit diagram of logarithmic amplifier and find its output voltage expression. 5
- (b) Find the input-output relation of the circuit shown in Fig. 4. 6
- (c) Find out the output voltage V_{OUT} of the circuit shown in Fig. 5. $A_f = \frac{A}{1+A\beta}$ 4
6. (a) Show that the overall gain of the negative feedback amplifier is $A_f = \frac{A}{1+A\beta}$. 4
- (b) Discuss various types of feedback mechanisms with the help of appropriate block diagram. 4+4 = 8
- (c) Estimate the input impedance Z_{if} and output impedance Z_{of} of voltage-series feedback. 2+5 = 7
7. (a) Draw and explain the operation of mono-stable multivibrator using IC-555 timer with waveform diagram. 4
- (b) Show that the duration of the pulse is $T = 1.1 RC$. 5
- (c) Calculate the frequency of oscillation of astable multivibrator using IC-555 timer having parameters $R_A = 4.7k\Omega$, $R_B = 3.3k\Omega$ and $C = 0.01\mu F$. 5
- OR
7. (a) State Barkhausen criteria for oscillation and explain the working principle of Wien bridge oscillator. 8
- (b) Explain operation of astable multivibrator using IC 555 timer in-detail and also write down the expression of frequency in output waveform. 8

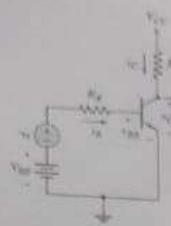


Fig. 1

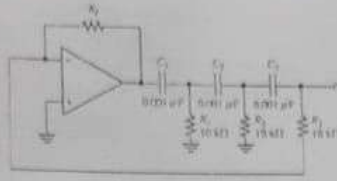


Fig. 2

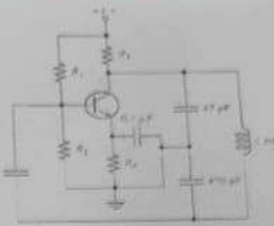


Fig. 3

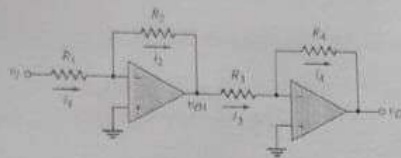


Fig. 4

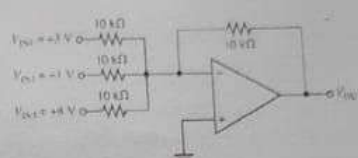


Fig. 5

8. (a) Draw and explain the operation of square-wave generator using OPAMP with waveform diagram. 1+5
 - (b) Derive the expression for the time period of output waveform. 7
 - (c) How the amplitude of the square-wave can be adjusted with back to back zener diodes. 3
- OR
8. Write short notes on any two of the following: 8+8
 - (a) Monostable multivibrator.
 - (b) Square wave generation using OP-AMP.
 - (c) Emitter Follower.
9. (a) With help of appropriate circuit diagrams explain active low pass filter, active high pass filter, band pass and band stop filter. 10
 - (b) For an emitter-bias configuration-based circuit the following specifications are given such as $I_{CQ} = (1/2) I_{sat}$, $I_{sat} = 8 \text{ mA}$, $V_{CC} = 28 \text{ V}$, $V_C = 18 \text{ V}$ and $\beta = 110$. Determine R_C , R_E and R_B . (Draw the appropriate circuit diagram). 6
10. (a) Draw and explain the Voltage divider bias circuit arrangement of an n -channel JFET in common source configuration. 8
 - (b) An n -channel JFET with voltage divider biasing scheme has the following device parameters such as $V_P = -5 \text{ V}$, $I_{DSS} = 12 \text{ mA}$, $R_1 = 400 \text{ k}\Omega$, $R_2 = 90 \text{ k}\Omega$, $R_D = 2 \text{ k}\Omega$, $R_S = 2 \text{ k}\Omega$ and $V_{DD} = 18 \text{ V}$. Find the operating point. 8

ALIAH UNIVERSITY
 ODD SEMESTER EXAMINATION, AUTUMN-2024
 PAPER: MATUGBS03 (Engineering Mathematics III)

Full Marks: 80, Time: 3 Hours

Use different answer scripts for each unit, All symbols have their usual meanings

Unit-I

Answer any EIGHT from the following questions:

8 × 5 = 40

- ✓ 1. Solve the following partial differential equation by Lagrange's method 5

$$z(x+y)p + z(x-y)q = x^2 + y^2$$

- ✓ 2. Solve the following partial differential equation by Lagrange's method 5

$$(3z - 5y)p + (5x - 4z)q = 4y - 3x$$

- ✓ 3. Solve the following non-linear partial differential equation by Charpit's method 5

$$z^2(p^2z^2 + q^2) = 1$$

- ✓ 4. Solve the following non-linear partial differential equation by Charpit's method 5

$$2zx - px^2 - 2qxy + pq = 0$$

- ✓ 5. Eliminate the constants a and b to form the PDE 3+2

$$(a)z = (x-a)^2 + (y-b)^2; \quad (b)z = ax + by + ab$$

6. Eliminate the function ϕ to form the PDE 5

$$\phi(x+y+z, x^2+y^2+z^2) = 0$$

- ✓ 7. Classify the following PDEs 2+3

(a) $4u_{xx} + 6u_{xy} + 9u_{yy} + 100u_x - 100u_y = 0$

(b) $2u_{xx} + 4u_{xy} + 3u_{yy} - 10u_x - 7u_y = 2$

- ✓ 8. Find the inverse of the Laplace transformation 5

$$L^{-1} \left[\frac{p^2 - 3p + 4}{p^3} \right]$$

- ✓ 9. (i) Find degree and order of the PDE 2+3

$$\frac{\partial^2 z}{\partial x^2} = \left(1 + \frac{\partial z}{\partial y} \right)^{\frac{1}{2}}$$

- (ii) Form the partial differential equation by eliminating the function f from the equation

$$y = f(x-at) + f(x+at)$$

10. Find the Laplace transformation of the following functions

(a) $f(t) = e^{-2t} \cos 3t$, (b) $g(t) = \sin^2 t - t$.

11. Find the Fourier series for $f(x) = |\cos x|$ in $(-\pi, \pi)$ of periodicity 2π .

12. Derive the Fourier series of $f(x) = x + x^2$ in $(-\pi, \pi)$ of periodicity 2π and hence deduce $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$.

Unit-II

Answer any EIGHT from the following questions:

8 × 5 = 40

1. What does the equation

$$x^2 + 4xy + y^2 - 2x + 2y + 6 = 0$$

become when origin moved to the point $(-1, 1)$ and the axes are turned through an angle of 45° in anticlockwise direction.

2. Prove that the equation to the straight lines through the origin each of which makes an angle α with the straight line $y = x$ is $x^2 - 2xy \sec 2\alpha + y^2 = 0$.

3. Prove that the equation $x^2 + 6xy + 9y^2 + 3x + 9y = 10$ represents a pair of parallel straight lines and find the distance between them.

4. Prove that the straight line $px + qy + r = 0$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $r^2 = a^2 p^2 + b^2 q^2$.

5. If the pole of the straight line with respect to the circle $x^2 + y^2 = c^2$ lies on $x^2 + y^2 = k^2 a^2$, then prove that the straight line will touch the circle $x^2 + y^2 = \frac{a^2}{k^2}$.

6. Reduce the following equations to their canonical forms and determine the nature of the conic represented by the equation $3x^2 + 2xy + 3y^2 - 16x + 20 = 0$.

7. Show that the pair of straight lines whose direction cosines are given by $3lm - 4ln + mn = 0$ and $l + 2m + 3n = 0$ are at right angles.

8. (a) Find the equation of the plane parallel to the plane $2x - 2y - z = 3$ and situated at a distance 7 unit from it.

(b) If a straight line makes an angle $\frac{\pi}{3}$ with x -axis, $\frac{\pi}{4}$ with y -axis and $\theta \in [0, \pi]$ with z -axis. Then find the value of θ .

9. Find the equation of the plane which passes through the point $(2, 1, -1)$ and orthogonal to each of the planes $x - y + z = 1$ and $3x + 4y - 2z = 0$.

10. Find the image of the point $(-3, 8, 4)$ in the plane $6x - 3y - 2z + 1 = 0$.

11. Find the shortest distance between the straight lines

$$\frac{x}{1} = \frac{y-1}{0} = \frac{z-1}{1} \text{ and } \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}.$$

12. Find the centre and the radius of the circle

$$x^2 + y^2 + z^2 = 25, x + 2y + 2z + 12 = 0.$$

Aliah University
Theoretical Examination, 2024
UG section (Science & Engg/Autumn Semester)
Sub: Indian Constitution (UCCUGAU03)

Full Marks: 80

Time: 3hrs

1. Write any ten questions: (within 30 words each)

2x10=20

- a. Which principles were added to the Constitution of India by the 42nd Amendment Act?
- b. Discuss Article 17 of the Constitution of India.
- c. Discuss Article 21A of the Constitution of India.
- d. Explain the term of 'Prohibition'.
- e. Write down any two fundamental duties mentioned in our Constitution.
- f. What is the composition of the Electoral College that elects the President?
- g. What is 'Ordinance'?
- h. What do you mean by the impeachment of the President of India?
- i. Discuss any two functions of the Speaker of the Lok Sabha.
- j. Mention the qualifications of the members of the Lok Sabha.
- k. Discuss the composition of the Upper House of the Indian Parliament.
- l. Mention two areas where the Rajya Sabha is more powerful than the Lok Sabha.

2. Write any four questions: (within 100 words each)

5x4=20

- a. Write a short note on the writs of Habeas Corpus and Quo Warranto.
- b. Critically discuss the Right to Freedom of Religion.
- c. Enumerate the differences between the Fundamental Rights and the Directive Principles of State Policy.
- d. Describe the election procedure for the President of India.
- e. Mention Five Fundamental Duties mentioned in our Constitution.
- f. Write any five Directive Principles of the State Policy.

2. Write any four questions: (Within 350 words each)

10x4= 40

- a. Discuss the basic features of the Indian Constitution.
- b. Discuss articles 14 to 18 of the Indian Constitution.
- c. Discuss Article 19 as enumerated in the Constitution of India and point out its reasonable restrictions.
- d. "The Constitution of India is neither purely federal nor unitary but a combination of both". Do you agree with this view? Give reasons in support of your answer.
- e. Discuss the powers and functions of the Prime Minister of India.
- f. Analyse the relationship between the two houses of Parliament.

B.Tech. Examination-2023-24
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)
Course Title: SIGNALS AND NETWORKS Course Code: ECEUGPC02

Full Marks: 80

Time: 3.00 Hrs

- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols have their usual meaning

Answer Question No. 1 and any 5 from Question No. 2 to 8

Q. Questions

No. Answer any *ten* from the following question: $10 \times 1 = 10$

1. (i) The differentiation of unit step signal is
 - (a) ramp signal
 - (b) impulse signal
 - (c) exponential signal
 - (d) parabolic signal
- (ii) If $x(t)$ is a continuous signal and $\delta(t)$ is a unit impulse signal then,

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)dt$$
 is equal to,
 - (a) $x(t)$
 - (b) $\delta(t)$
 - (c) $x(t_0)$
 - (d) $\delta(t_0)$
- (iii) Which of the following statements are true?
 - (a) an LTI system is always stable system
 - (b) an LTI system is stable only if the integral of its impulse response is finite
 - (c) in a system, if the input is bounded then the output is also bounded
 - (d) in a system, even if the input is bounded the output can be bounded
- (iv) ROC of a causal signal $x(t)$ is
 - a) Entire s-plane
 - b) Right half of s-plane
 - c) Left half of s-plane
 - d) None
- (v) Given that $H(s) = \exp(-4s)$. What is the impulse response of the system?
 - (a) $\delta(t-4)$
 - (b) $u(t-4)$
 - (c) $\exp(-4t)u(t)$
 - (d) $\exp(4t)u(t)$

$H(s) = e^{-4s}$
 $h(t) = \delta(t-4)$
- (vi) If Fourier transform of $x(t)$ is $X(\Omega)$ then Fourier transform of $\exp(-2j\omega_0 t)x(t)$ is
 - (a) $X(\Omega)/2\omega_0$
 - (b) $X(\Omega+2\omega_0)$
 - (c) $X(\Omega-\omega_0)$
 - (d) $X(\Omega+\omega_0)$
- (vii) Which of the following signal is a periodic signal
 - (a) $x(t) = Au(t)$
 - (b) $x(t) = Ae^{j\omega t}$
 - (c) $x(t) = Ae^{bt}$
 - (d) $x(t) = At$
- (viii) How can we define half wave symmetry?
 - a) $x(t) = -x(t \pm T)$
 - b) $x(t) = x(t \pm T/2)$
 - c) $x(t) = x(t \pm T)$
 - d) $x(t) = -x(t \pm T/2)$

$x(t) = x(t+T)$
 $e^{-j\omega T} = 1$

(ix) Find the Fourier transform of $x(t) = f(t-2) + f(t+2)$.

- a) $2F(\omega)\cos 2\omega$
- b) $F(\omega)\cos 2\omega$
- c) $2F(\omega)\sin 2\omega$
- d) $F(\omega)\sin 2\omega$

(x) Fourier series of a signal gives following output:

- a) Continuous-time and Periodic
- b) Discrete-time and Aperiodic
- c) Discrete-time and Periodic
- d) Continuous-time and Aperiodic

(xi) Determine the odd component of the signal; $x(t) = \cos t + \sin t$

- a) $\sin t$
- b) $2\sin t$
- c) $\cos t$
- d) $2\cos t$

$$\begin{aligned} x(-t) &= -\cos t - \sin t \\ x(t) &= \cos t + \sin t \\ x(-t) &= \cos(-t) + \sin(-t) \end{aligned}$$

(xii) Which of the passive component performs differentiation operation?

- a) Resistor
- b) Capacitor
- c) Inductor
- d) Amplifier

Q2. (a) Given a signal $x(t) = \begin{cases} 1/6(t+2), & -2 \leq t \leq 4, \\ 0, & \text{otherwise} \end{cases}$

8

Then Sketch (i) $x(t)$ (ii) $x(t+1)$ (iii) $x(2t)$ (iv) $x(t/2)$

(b) Find the energy spectral density of the signal $x(t) = 10 \text{ Sinc } 10t$. Also find its total energy.

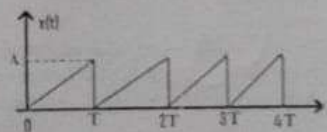
6

3. (a) Define Laplace transform of signal $x(t)$ and its region of convergence

3

(b) Find out the Laplace transform of the signal shown in below figure.

5



$$\begin{aligned} (t, A) \\ (0, 0) \\ \frac{y-A}{x-c} &= \frac{e^{-T/s}}{s} \\ y &= A \end{aligned}$$

(c) State and prove initial value and final value theorems of Laplace transform.

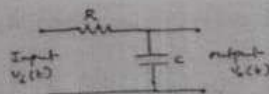
6

4. (a) Prove that output of an LTI system is the convolution sum of input $x(t)$ and impulse response $h(t)$, i.e. $y(t) = x(t) * h(t)$

6

(b) Find the transfer function and impulse response of the following RC circuit:

8



5. (a) A causal LTI system is described by $y[n] - y[n-1] + y[n-2] = x[n]$, where $x[n]$ is the input to the system $h[n]$ is the impulse response of the system. Find System function $H(z)$ Impulse response $h[n]$.

7

(b) Determine the Nyquist sampling rate and Nyquist sampling interval for the signal $x(t) = \text{sinc}^2(200t)$

3

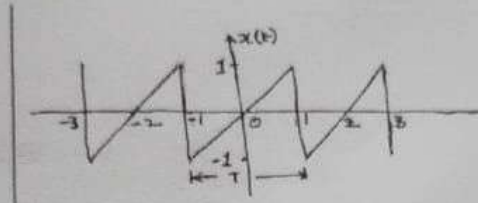
(c) Prove the following properties of Fourier transform

4

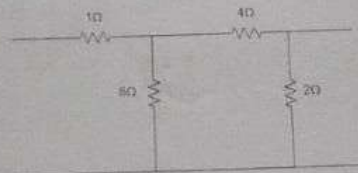
- i) Frequency shifting
- ii) Time differentiation

- (a) Find the trigonometric Fourier series for the periodic signal shown in Fig.

6



- (b) Obtain the Fourier transform of the following functions: i) Impulse function ii) DC signal iii) Unit step function. 6
- (c) What is aliasing? 2
7. (a) Determine the response of an LTI system whose impulse response $h(n)$ and $x(n)$ are given by $h(n)=[1, 2, 1, -2, -1]$, $x(n)=[1, 2, 3, -1, -3]$ 8
- (b) Determine the response of discrete time system governed by the difference equation, $y(n) = -0.5y(n-1) + x(n)$ when the input is unit step, and with initial condition a) $y(-1)=0$. 6
8. (a) Express the two port network in terms of short circuit admittance $[Y]$ parameter. 4
- (b) What is symmetry and reciprocity condition of Y parameter, prove them. 4
- (c) For the given network, find the open-circuit impedance parameters. Also draw its equivalent circuit. 6



-End-